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UNIT-4

EMERGING TRENDS (ET)

In this chapter

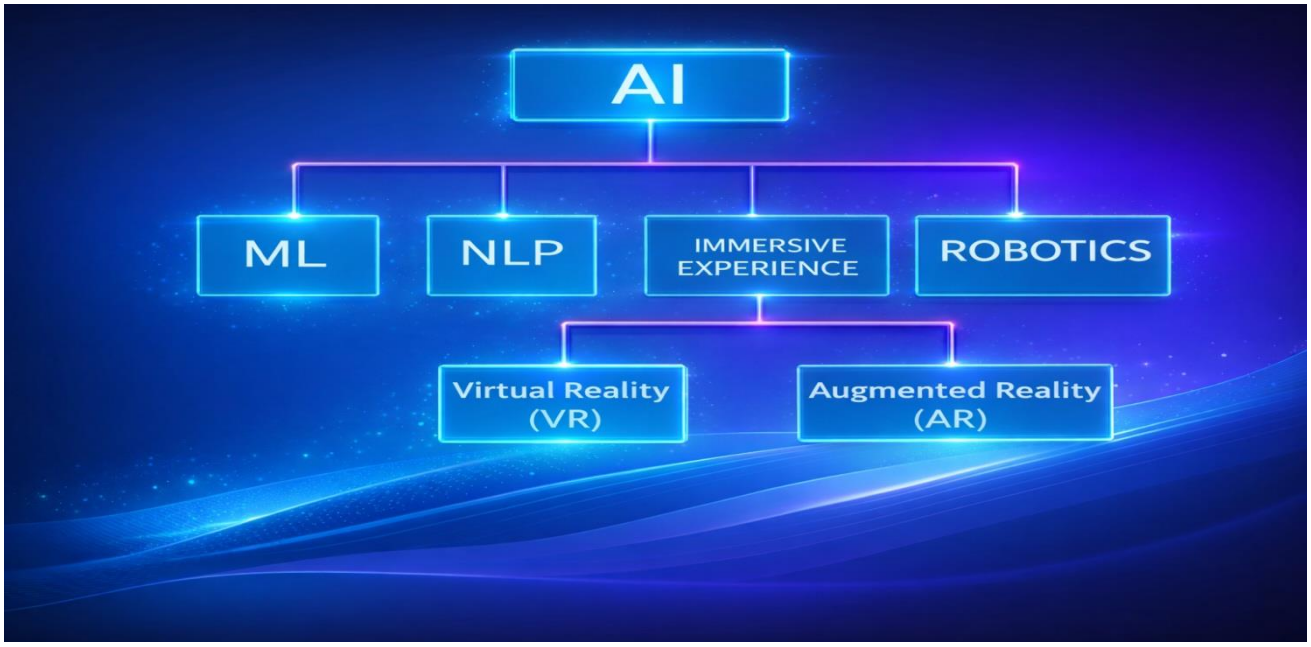
- Introduction to Emerging Trends
- Artificial Intelligence (AI)
- Big Data
- Internet of Things (IoT)
- Cloud Computing
- Grid Computing
- Blockchains

Unit 4: Introduction to the Emerging Trends:

- Artificial Intelligence,
- Machine Learning,
- Natural Language Processing,
- Immersive experience (AR, VR),
- Robotics, Big data, and its characteristics,
- Internet of Things (IoT),
- Sensors, Smart Cities,
- Cloud Computing and Cloud Services (SaaS, IaaS, PaaS);
- Grid Computing,
- Blockchain technology.

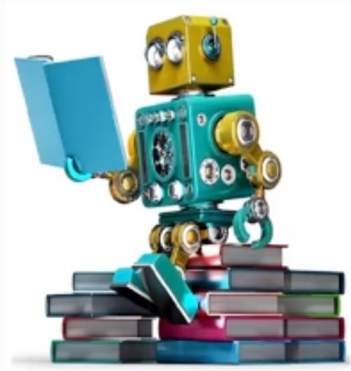
1. **Introduction to Emerging Trends:** Emerging trends are the state-of-the-art technologies (Latest Technologies) which gain popularity and set a new trend among users.

2. **Artificial Intelligence**: Artificial Intelligence refers to technologies that help build machines that can **think**, **act, decision making, problem solving** and **learn** like humans.
Artificial Intelligence aims to simulate natural intelligence of humans into machines.
John McCarthy is known as father of AI.
Ex: Self Driving,Digital Assistants,Google Maps,Face Recognition,Online Ads



2.2.1 Machine Learning: Machine learning is an **subset/application** of (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. Machine learning algorithms (models) use historical data as input to predict new output values.

Ex: Gmail spam filter,YouTube/Netflix Recommendations, Online Shopping Recommendations



2.2.2 Natural Language Processing (NLP): It is a branch of artificial intelligence that deals with the interaction between computers and humans using the natural language. The ultimate objective of NLP is to read, understand, and make sense of the human languages

Some examples of NLP are

a) The predictive typing feature in a search engine. In Google, when a word is typed then the next suggestive word will be shown automatically

b) The Text–to–Speech and Speech–to–Text features of websites or machines

c) Machine translation from one language to another language such as English to Hindi etc

d) An automated customer service where computer software can interact with customers to serve their queries of complaints.

2.2.3 Immersive Experience: It allows us to **visualize**, **feel** and **react** by stimulating our senses. Immersive experiences have been used in the field of training, such as driving & flight simulator and so on. Immersive experience can be achieved using **Virtual Reality** and **Augmented Reality**.

A) Virtual Reality: It is a **three-dimensional, computer–generated situation** that simulates the real world. The user can interact with and explore that environment by being immersed in it while interacting with the objects and other actions of the user.

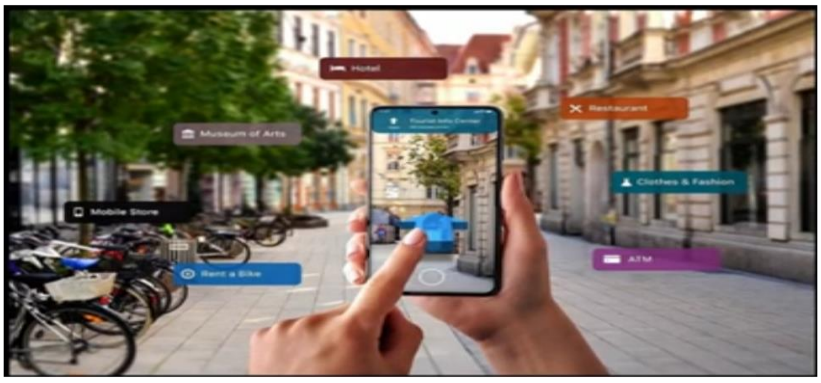
At present, it is achieved with the help of **VR Headsets**. It has its applications in gaming, military training, medical procedures, entertainment, social science and psychology, engineering & other areas where simulation is needed for a better understanding

Ex: https://science.nasa.gov/eyes/, Google Earth



B) Augmented Reality: The superimposition of computer generated perceptual information over the existing physical surroundings is called as **Augmented Reality (AR).** It adds components of the **digital world to the physical world**. Users can access information about the nearest places with reference to their current location. With the help of **location–based AR App**, travellers can access real–time information of historical places just by pointing their camera to the place.AR can be accessed with a **smartphone**.

Example : Snapchat,Angry Birds, Pokémon Go, Interior decoration apps,Google glass, Google street view,Microsoft Hololens,IKEA app,Instagram filter(Sensors:Accelerometer,Gyroscopes)





2.2.4 Robotics: A robot is a machine capable of carrying out one or more tasks automatically with accuracy and care. A robot is programmable, which means it can follow the instructions given through computer programs.

Robots were initially conceptualized for doing repetitive industrial tasks that are boring or stressful for humans. **Sensors** are one of the prime components of a robot.

Robot can be of many types, such as **wheeled robots, legged robots, manipulators and humanoids**. Robots that resemble humans are known as humanoids.

Robots are being used in industries, medical science, bionics, scientific research, military, etc.

Some examples are:

- NASA's **Mars Exploration Rover** (MER) mission is a robotic space mission to study about the planet Mars



- Sophia is a humanoid that uses artificial intelligence, visual data processing, facial recognition and also imitates human gestures and facial expressions



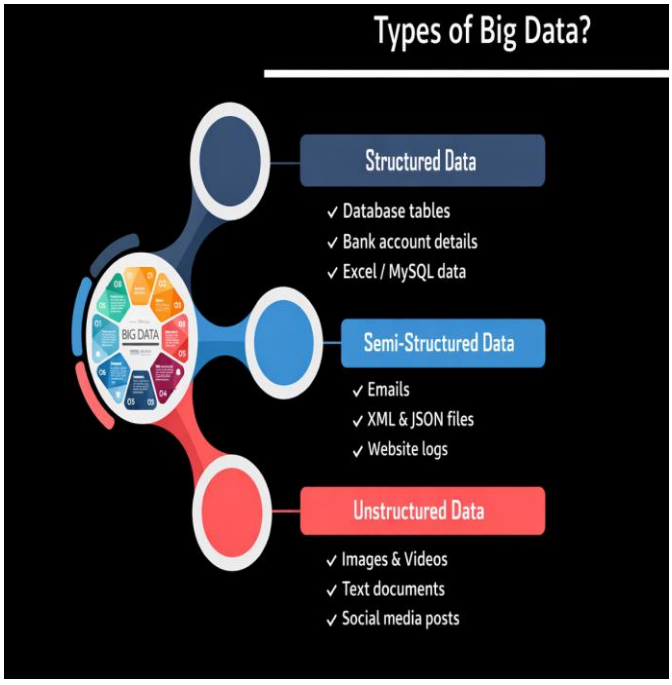
- A drone is also called as **Quadcopter or (UAV -Unmanned Aerial Vehicle)** is an aircraft, which can be remotely controlled or can fly autonomously through software–controlled flight plans in their embedded systems, working in conjunction with onboard sensors and GPS.



These are used in journalism, filming and aerial photography, shipping or delivery at short distances, disaster management, search and rescue operations, healthcare, geographic mapping and structural safety inspections, agriculture, wildlife monitoring, law–enforcement and border patrolling.

- **BIG DATA:** Big Data is a huge amount of data generated at very high speed and in different formats, which requires special technologies to store and process.
- Big data not only represents voluminous data, it also involves various challenges like **integration, storage, analysis, searching, processing, transfer, querying and visualisation** of such data Ex: Covid Analysis
- Due to increase in internet, smart phone users and world's web traffic huge quantity of data is being generated every day. The data is not only voluminous, but also unstructured like posts, instant messages, audio / video chats, photographs, tweets, blog articles, news items, opinion polls etc. This is called Big Data. Such data cannot be processed and analysed using traditional data processing tools.





Characteristics of Big Data: Big data exhibits the following **five** characteristics that distinguish from traditional data .



i. **Volume**: The most prominent characteristic of big data is its **enormous size**. If a particular data set is of such large size that it is difficult to process it with traditional DBMS tools, it can be termed as big data. Ex: **KB,MB,GB,TB,PB,EB,ZB,YB**

ii.Velocity: It refers to the **speed** at which data is generated and the rate at which it must be processed. Ex**: YouTube, Instagram, Facebook,X**

iii.Variety: It refer to the different types of data generated, such as **structured**, **semi–structured** and **unstructured data**. Ex:**Text, images, videos, web pages and so on**.

iv.Veracity: Veracity refers to the trustworthiness of the data because processing such incorrect data can give wrong results or mislead the interpretations**. Ex:Trust or Fake**

v. Value: Big data is not only just a big quantity of data, but also possess to have hidden patterns and useful knowledge, which can be of high business value. Ex: **Reel or JEE**

Ex:Market or Business
Volume-Size(Enormous size Data)
Velocity-Speed(Youtube)
Variety-Stuctured,Unstructured,Semi Structured
Veracity-Trust or Fake
Value-market

Data Analytics: Data analytics is the process of examining data sets in order to draw conclusions about the information they contain, by using specialised systems and software.

Data analytics technologies and techniques are becoming popular. They are used in commercial industries to enable organisations to make more informed business decisions. In the field of science and technology, it can be useful for researchers to verify or disprove scientific models, theories and hypotheses.

Pandas is a library of the programming language Python that can be used as a tool to make data analysis much simpler.

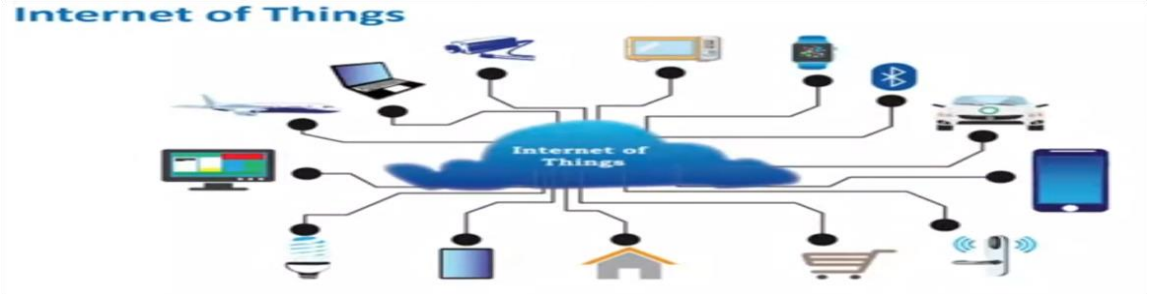


2.4 Internet of Things (IoT):

Internet of Things (IoT) is a system where physical devices embedded with sensors, hardware and software are connected to the internet so they can collect, share, and act on data automatically.**Kevin Ashton** is known as the **Father of the Internet of Things** (IoT).

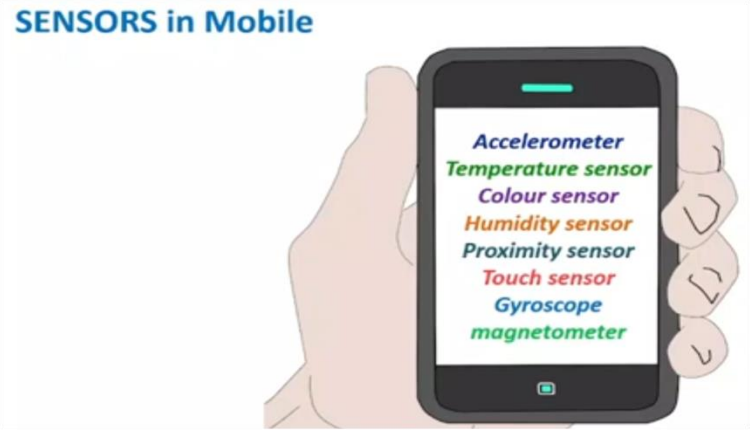
In a typical household, many devices have advanced hardware and software. These devices are used in isolation from each other.IoT tends to bring together these devices to work in collaboration and assist each other in creating an intelligent network of things.

Ex: microwave oven, an air conditioner, door lock, CCTV camera or other such devices are enabled to connect to the Internet, then these devices can be accessed and remotely controlled using a smart–phone or computer.



2.4.1 Web of Things (WoT): Web of Things (WoT) allows the use of **web services** to connect anything in the physical world, besides human identities on web. It will pave way for creating smart homes, smart offices, smart cities and so on.

2.4.2 Sensors: A smart sensor is a device that takes input from the physical environment and uses built-in computing resources to perform predefined functions upon detection of specific input and then process data before passing it on. The evolution of smart electronic sensors is contributing in a large way to the evolution of IoT.

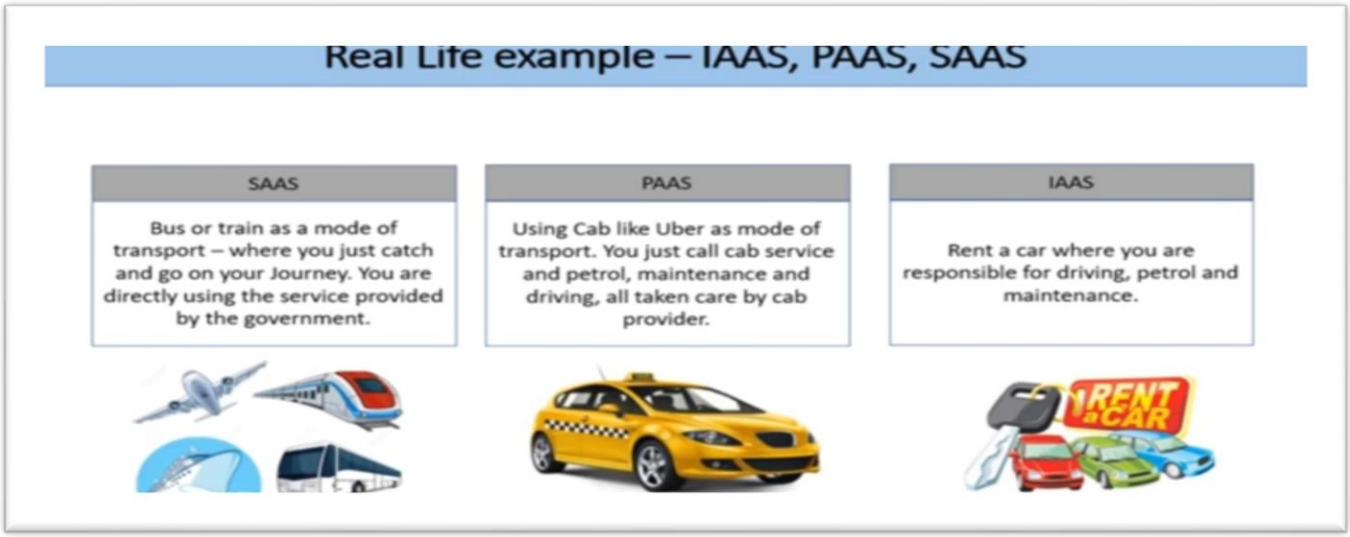


2.4.3 Smart Cities :A smart city is an urban area that uses different types of electronic Internet of things (IoT) sensors to collect data and then use insights gained from that data to manage assets, resources and services efficiently. Smart city applications are developed to manage urban flows and allow for real-time responses.

Ex: A **smart building** uses sensors to detect earthquake and alerts. A **smart bridge** uses wireless sensors to detect any loose bolt, cable or crack and alerts concerned authorities through SMS. A **smart tunnel** uses wireless sensors to detect any leakage or congestion in the tunnel and the information can be sent to a centralized computer for further analysis.

2.5 Cloud Computing

Cloud computing means storing and accessing data and programs **over the Internet** or the **cloud**, The services comprise software, hardware (servers), databases, storage, etc. These resources are provided by companies called **cloud service providers** and usually charge on pay per use basis.



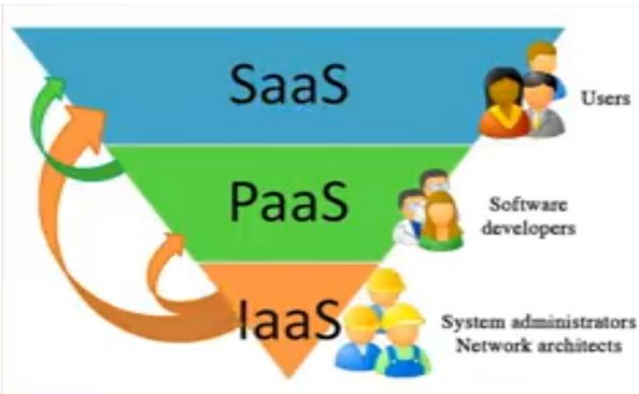
2.5.1 Cloud Services: There are three standard models to categorise different computing services. These are **Software as a Service (SaaS), Platform as a Service (PaaS), Infrastructure as a Service (IaaS**)

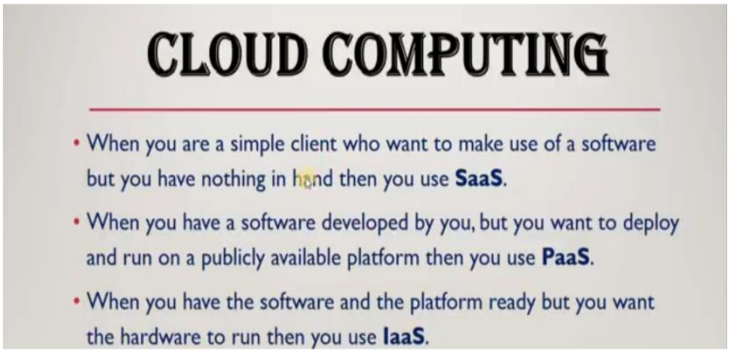
(A) Software as a Service (SaaS) :SaaS provides on-demand access to application software, usually requiring a licensing or subscription by the user. Ex:Google drive,Office 365,Netflix

(B) Platform as a Service (PaaS) : PaaS provides a platform or environment to develop, test, and deliver software applications. Ex: Microsoft Azure

(C) Infrastructure as a Service (IaaS),: The IaaS providers can offer different kinds of computing infrastructure, such as **servers, virtual machines (VM), storage and backup facility, network components**, operating systems or any other hardware or software.

Ex:**Top 5 cloud service Providers**:1.Amazon Web Services(AWS), 2.Microsoft Azure,
3.Google Cloud Platform (GCP), 4.IBM Cloud, 5.Oralce Cloud





Grid Computing: A Grid is a computer network of geographically dispersed and heterogeneous computational resources. A grid is more application specific and creates a sense of a virtual supercomputer with an enormous processing power and storage.

Or

Grid computing is a computing method that uses multiple computers to work together to solve complex problems. It's a type of distributed computing that uses a network to make a group of computers appear as a single computing entity.

Grid can be of **two** types —

a.Data grid, used to manage large and distributed data having the required multi-user access

b.CPU or Processor grid, where processing is moved from one PC to another as needed or a large task is divided into subtasks, and allotted to various nodes for parallel processing.

Cloud Computing	**Grid Computing:**
Cloud Computing follows client-server computing architecture	Grid computing follows a distributed computing architecture
Cloud Computing is more flexible.	Grid Computing is less flexible.
Cloud operates as a centralized management system	Grid operates as a decentralized management system
In cloud computing, cloud servers are owned by infrastructure providers.	In Grid computing, grids are owned and managed by the organization
Cloud computing uses services like - Iaas, PaaS, and SaaS.	Grid computing uses systems like distributed computing, distributed information, and distributed pervasive.
Cloud Computing is Service-oriented	Grid Computing is Application-oriented.

Blockchains: The blockchain technology works on the concept of three pillars a.**decentralised** b.**Transparency** and c. **immutability** of shared database where each computer has a copy of the database

Or

Blockchain is a system that allows a group of connected computers to maintain a single updated and secure ledger

Or

A block can be thought as a secured chunk of data or valid transaction. Each block has some data called its header, which is visible to every other node, while only the owner has access to the private data of the block. Such blocks form a chain called Blockchain.

The most popular application of blockchains technology is in digital currency. However, due to its decentralised nature with openness and security, blockchains are being seen as one of the ways to ensure transparency, accountability and efficiency in business as well as in governance systems.

1.What are the pillars of blockchain technology?
a.Transparency b.Immutability c.Decentralization d.**All of them**
2.What are the benefits of blockchain technology?
a.No hidden fees b.Fraud control & Access levels c.Security & Speed **d.All of the Above**.
3. What is a blockchain?
a.A blockchain is a centralized digital ledger consisting of records called blocks.
b.A blockchain is a decentralized, distributed, digital ledger consisting of records called blocks.
c.A blockchain is a digital database consisting of records called class.
d.It is a private ledger that no one can inspect.
4. Which technology is used in Bitcoin?
a.Information Technology b.Business Technology
c.Blockchain Technology d.Medical Technology